

SKILL WEEK 6

Task 1:

Code:

```
#include <stdio.h>

int main() {
    char input[200];
    int i;

    printf("Enter a string with escape characters: ");
    fgets(input, sizeof(input), stdin);

    printf("\nProcessed string:\n");

    for (i = 0; input[i] != '\0'; i++) {

        if (input[i] == '\\' && input[i + 1] != '\0') {

            i++;

            switch (input[i]) {

                case 'n':
                    printf("\n");
                    break;

                case 't':
                    printf("\t");
                    break;

                case ' ':
                    printf(" ");
                    break;

                case '\\':
                    printf("\\");
                    break;

                case '$':
```

```

        printf("$");
        break;

    case '!':
        printf("!");
        break;

    default:
        printf("%c", input[i]);
    }

} else {
    printf("%c", input[i]);
}
}

return 0;
}

```

Task 2:

Code:

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {

    pid_t pid;

    pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {

        // Child process
        printf("Child process created.\n");
        printf("Child PID: %d\n", getpid());

        char *args[] = {"ls", "-l", NULL};
    }
}

```

```

    execlvp(args[0], args);

    // This runs only if execlvp fails
    perror("execlvp failed");
    exit(1);

} else {

    // Parent process
    printf("Parent process.\n");
    printf("Parent PID: %d\n", getpid());

    wait(NULL);

    printf("Child process finished.\n");
}

return 0;
}

```

Output:

```

ments/Documents/rishish-MacBook-Air/Collage/CSS/Programs/Skill week 6 %
[rishivarma@rishish-MacBook-Air Skill week 6 % nano escape.c
[rishivarma@rishish-MacBook-Air Skill week 6 % gcc escape.c -o escape
[rishivarma@rishish-MacBook-Air Skill week 6 % ./escape
Enter a string with escape characters: hello\ world

Processed string:
hello world
[rishivarma@rishish-MacBook-Air Skill week 6 % ./escape
Enter a string with escape characters: hello\nworld

Processed string:
hello
world
[rishivarma@rishish-MacBook-Air Skill week 6 % nano child_exec.c
[rishivarma@rishish-MacBook-Air Skill week 6 % gcc child_exec.c -o child_exec
[rishivarma@rishish-MacBook-Air Skill week 6 % ./child_exec
Parent process.
Parent PID: 34254
Child process created.
Child PID: 34255
total 160
-rwxr-xr-x  1 rishivarma  staff   33816  21 Sep 09:25 child_exec
-rw-r--r--  1 rishivarma  staff    726  21 Sep 09:25 child_exec.c
-rwxr-xr-x  1 rishivarma  staff   33640  21 Sep 09:24 escape
-rw-r--r--  1 rishivarma  staff   1046  21 Sep 09:23 escape.c
Child process finished.
[rishivarma@rishish-MacBook-Air Skill week 6 % █

```